

GEOMETRY-AGNOSTIC ACOUSTIC PROCESSING: A DYNAMIC SPATIAL NETWORK FOR JOINT ECHO CANCELLATION AND NOISE SUPPRESSION

Kangqi Jing^{1†}, Wenbin Zhang^{3†}, Jun Du², Qing Wang^{2*}, Yu Gao^{3*}

¹School of Information Science and Engineering, Southeast University, China

²University of Science and Technology of China (USTC), Hefei, China

³AI Research Center, Midea Group (Shanghai) Co.,Ltd., China

jing kangqi@seu.edu.cn, zhangwb87@midea.com

ABSTRACT

We propose a geometry-agnostic solution for multichannel acoustic echo cancellation and noise suppression. The model employs a Dynamic Spatial Encoder that adaptively processes spatial information from arbitrary microphone arrays, combined with powerful separation modules for echo cancellation and noise suppression. A post-processing network further enhances output quality by eliminating residual artifacts and speech distortions. The system incorporates an integrated classifier for frame-level voice activity detection, enabling accurate identification of acoustic states (double-talk, far-end single-talk, near-end single-talk and double-silence). Experimental results demonstrate consistent performance across various array geometries for both stationary and moving speakers, achieving significant word error rate (WER) reduction in challenging conditions while improving perceptual speech quality.

Index Terms— Acoustic echo cancellation, multichannel, geometry-agnostic

1. INTRODUCTION

Acoustic echo cancellation (AEC) addresses the critical challenge of removing undesired echoes caused by loudspeaker-microphone coupling in communication systems [1]. Modern conference rooms, typically equipped with multiple microphones and loudspeakers, require multichannel solutions (MCAEC) and multi-loudspeaker echo cancellation (MMAEC) approaches. These scenarios present both additional challenges and opportunities compared to single-channel AEC, driving significant research attention [2].

While neural network-based multichannel AEC methods [3, 4, 5, 6] have demonstrated superior performance over conventional approaches [7, 8, 9], their real-world deployment faces a critical limitation: current models require retraining for different microphone array geometries. This constraint

necessitates impractical computational resources when adapting to diverse conference room configurations. Recent array-agnostic speech enhancement methods [10, 11, 12, 13, 14] have explored this direction. DeFTAN-AA [12] introduces a spatial transformer that aggregates spatial information through channel-wise self-attention and a space-object cross-attention (SOCA) block to mitigate array-specific overfitting. Similarly, Zheng *et al.* [14] proposes a spatial filter bank to extract geometry-invariant features, processed by a two-stage conformer-based model. ArrayDPS [13] addresses the blind speech separation (BSS) problem through an unsupervised, array-agnostic framework that leverages diffusion posterior sampling to approximate room acoustics without requiring prior knowledge of array geometry.

Building upon these advances, we formulate AEC as a supervised speech separation task. Our framework introduces three key components: (1) a Dynamic Spatial Encoder that adaptively processes spatial information from arbitrary microphone arrays, (2) powerful separation modules for joint echo cancellation and noise suppression, and (3) a post-processing network that eliminates residual artifacts and speech distortions. The system incorporates an auxiliary classification network for frame-level acoustic state detection, identifying four distinct scenarios: double-talk, far-end single-talk, near-end single-talk and double-silence. This state information guides the post mask estimation network to optimize echo cancellation performance. By estimating near-end speech from microphone signals using far-end references as conditional information, our approach decouples system performance from specific spatial configurations.

Experiments show that the proposed method effectively suppresses both acoustic echoes and background noise, even under challenging conditions with nonlinear distortions for adverse SER and both static and moving near-end sources. Furthermore, downstream ASR evaluation shows significant WER reduction compared to baseline approaches, confirming the practical utility of our geometry-agnostic design. Code and audio examples are available at <https://github.com/jingkangqi/VSAEC>.

[†] Equal contribution. ^{*} Corresponding author.

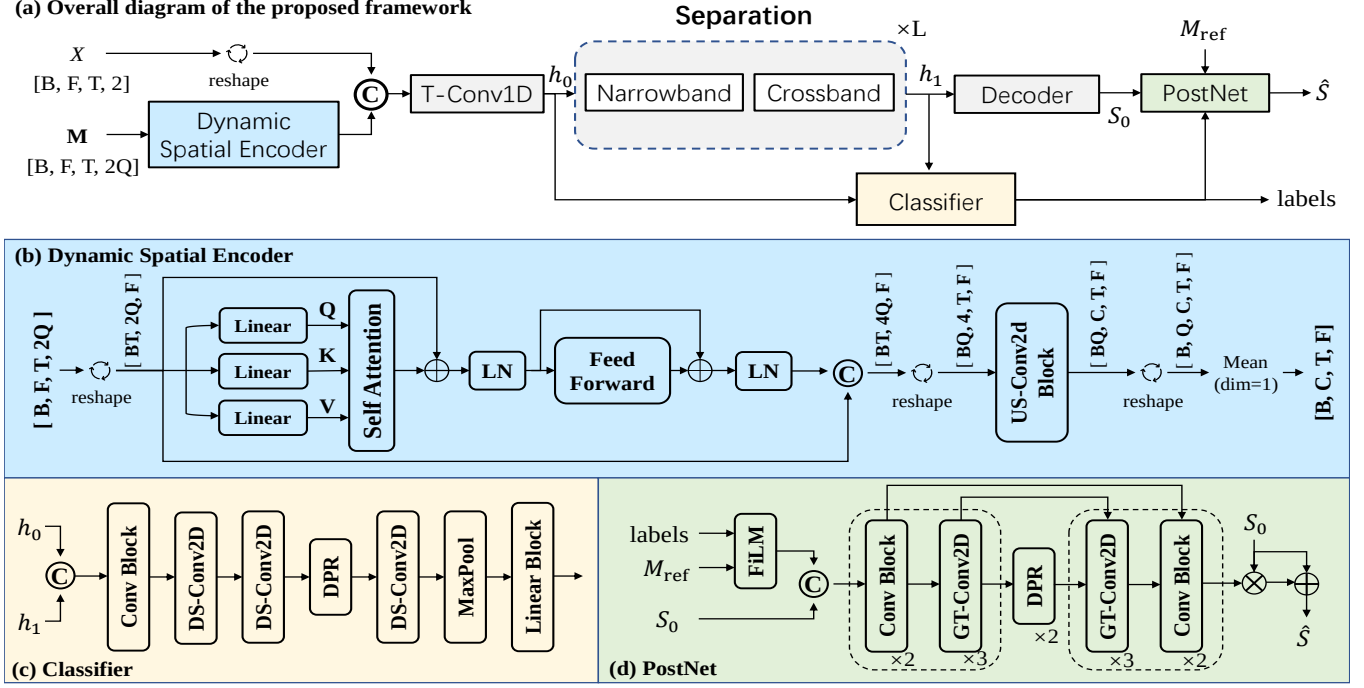


Fig. 1. (a) The system overview of the proposed AEC model. The detailed structure of (b) Dynamic Spatial Encoder, (c) Classifier, (d) PostNet.

2. PROBLEM FORMULATION

Consider a system comprising P loudspeakers and Q microphones. The loudspeaker's signal $x_p[n]$, generated from far-end sources $x[n]$, are convolved with room impulse responses (RIRs) $h_p[n]$, where $p \in \{1, 2, \dots, P\}$. In the receiving room, the processed signals $x_p^{nl}[n]$ are played through loudspeakers. Assume $s[n]$ is the clean near-end speech, the signal captured by the q -th microphone in the short-time Fourier transform (STFT) domain is expressed as:

$$M_q(t, f) = \sum_{p=1}^P \underbrace{H_{pq}(f)X_p^{nl}(t, f)}_{E_{pq}(t, f)} + S_q(t, f) + V_q(t, f), \quad (1)$$

where t and f denote the time frame and frequency bin indices respectively, $E_{pq}(t, f)$ represents the echo component from the p -th loudspeaker to the q -th microphone, $H_{pq}(f)$ is the corresponding acoustic transfer function, $S_q(t, f)$ denotes the STFT of the reverberant near-end speech signal, given by $s_q[n] = s[n] * r_q[n]$, $r_q[n]$ is the RIR between the clean near-end source and the q -th microphone, and $V_q(t, f)$ is the background noise. The multichannel mixture received by the Q -microphone array at each time-frequency bin is given by:

$$\mathbf{M}(t, f) = \mathbf{E}(t, f) + \mathbf{S}(t, f) + \mathbf{V}(t, f) \in \mathbb{C}^Q, \quad (2)$$

where $\mathbf{E}(t, f)$, $\mathbf{S}(t, f)$, and $\mathbf{V}(t, f)$ represent the echo, speech, and noise components respectively.

The objective of this work is to simultaneously suppress the echo $\mathbf{E}(t, f)$ and noise $\mathbf{V}(t, f)$, and to perform dereverberation to recover an enhanced speech signal $\hat{s}[n]$ such that $\hat{s}[n] \mapsto s[n]$. Our proposed system achieves this through a composite transformation $F(\cdot)$ operating on both the microphone signals and the reference far-end signal:

$$\hat{s}[n] = \text{iSTFT}(F(\mathbf{M}(t, f), X(t, f))), \quad (3)$$

where $X(t, f)$ represents the STFT of the reference channel's far-end signal $x[n]$. The system is designed to adapt flexibly to varying numbers of microphones (Q) and loudspeakers (P), arbitrary array geometries, while jointly addressing acoustic echo cancellation, noise suppression, and speech dereverberation in a unified framework.

3. PROPOSED METHOD

The proposed AEC system, illustrated in Figure 1, processes multi-microphone STFT coefficients $\mathbf{M} \in \mathbb{R}^{F \times T \times 2Q}$ and far-end reference signal $X \in \mathbb{R}^{F \times T \times 2}$ to estimate the clean signal $\hat{s}$, where F and T denote the number of frequency bins and time frames, respectively.

The model incorporates a Dynamic Spatial Encoder to capture spatial correlations between arbitrary microphone pairs, enabling channel-agnostic processing for variable array geometries. The encoded microphone features are concatenated with the far-end reference signal, and across the convolutional input layer (T-Conv1d) with a kernel size of 1 and

5 on the frequency and time axes, respectively, and with C channels, obtaining a hidden representation $h_0 \in \mathbb{R}^{F \times T \times C}$.

h_0 is then processed by interleaved Crossband and Narrowband layers to achieve simultaneous speech separation and echo cancellation. The Crossband and Narrowband layers follow the same structure as described in [15], selected for their strong ability to separate speech using complex spatial information. The output of this processing is $h_1 \in \mathbb{R}^{F \times T \times C}$.

The features h_0 and h_1 are concatenated and fed to the Classifier to predict the acoustic scenario labels. The decoder consists of a linear layer that maps h_0 to the preliminary target STFT coefficients S_0 . A postnet is then utilized to recover speech distortions and remove residual noise in the audio signal. Finally, inverse short-time Fourier transform (iSTFT) is applied to resynthesize the echo-canceled signal $\hat{s}$.

3.1. Dynamic Spatial Encoder

The proposed Dynamic Spatial Encoder, illustrated in Figure 1(b), introduces a geometry-agnostic framework for spatial feature extraction in microphone arrays by adaptively processing input features $\mathbf{X} \in \mathbb{R}^{F \times T \times 2Q}$ through a hierarchical spatial attention mechanism. The encoder first reinterprets the conventional feature dimensions by treating temporal frames (T) as batch dimension, frequency bins (F) as the embedding dimension, and microphone channels ($2Q$) as the sequence dimension, enabling the generation of a $2Q \times 2Q$ spatial attention matrix that adaptively captures pairwise microphone interactions independent of array configuration. The attended features are then enhanced through a feedforward network with layer normalization(LN)[16] to introduce nonlinear feature transformations. These processed features are then combined with the reshaped original spectral inputs, where the real and imaginary components are systematically separated and reorganized into a $(Q, 4, T, F)$ -dimensional representation through a learnable upsample convolutional (US-Conv2d) block consisting of a convolution layer, LN and parametric rectified linear unit (PReLU) activation[17]. Finally, a mean pooling along the microphone-dimension to generate consolidated features $\mathbf{O} \in \mathbb{R}^{B \times C \times T \times F}$ that preserve essential spatial information while remaining adaptable to arbitrary microphone configurations.

3.2. Classifier

The classifier, illustrated in Figure 1(c), provides frame-level acoustic scenario detection to guide the post-enhancement process. The network first projects the concatenated high-dimensional features $[h_0; h_1] \in \mathbb{R}^{F \times T \times 2C}$ into a lower-dimensional label space through a convolutional block, followed by two successive downsampling convolutional blocks (DS-Conv2D) that progressively reduce the frequency resolution to $F/4$, balancing computational efficiency with feature retention. At the bottleneck, a Dual-Path Recurrent (DPR)

module [18] simultaneously models local band-wise temporal dependencies and global frame-wise frequency correlations through its dual-path architecture, effectively capturing essential scenario characteristics with minimal parameters. Then a DS-Conv2D block projects features to dimension D . The final classification head consists of a max-pooling operation along the frequency dimension, a fully-connected layer with weight matrix $\mathbf{W}_c \in \mathbb{R}^{D \times N_{\text{class}}}$, and a softmax activation that generates the posterior probability distribution $\hat{p}_t \in [0, 1]^{N_{\text{class}}}$ for each frame, where t denotes the frame index and N_{class} denotes the number of predefined acoustic scenarios.

3.3. PostNet

The postnet serves as the final refinement stage for the enhanced spectrogram, addressing residual artifacts and improving speech quality. As shown in Figure 1(d), the module processes three inputs: the primary enhanced spectrogram $S_0 \in \mathbb{C}^{F \times T}$, scenario labels $\mathbf{L} \in \mathbb{R}^{T \times N_{\text{class}}}$ from the classifier, and the reference channel's microphone signal $M_{\text{ref}} \in \mathbb{C}^{F \times T}$.

The post-processing begins with FiLM-conditioned [19] feature transformation, where the reference signal is modulated according to acoustic scenarios through learned scaling and shifting parameters:

$$\text{FiLM}(M_{\text{ref}}|\mathbf{L}) = \gamma(\mathbf{L}) \odot M_{\text{ref}} + \beta(\mathbf{L}) \quad (4)$$

where $\mathbf{L} \in \mathbb{R}^{T \times N_{\text{class}}}$ represents the frame-wise scenario labels from the classifier, and γ, β are learned transformations according to the current state. The FiLM-processed features are then concatenated with S_0 along the channel dimension and fed into a U-Net architecture.

The encoder consists of two convolution block to map the input spectrum to a high-dimensional and three grouped temporal convolution (GT-Conv) blocks employing depth-wise separable convolutions with dilation rates (1, 2, 5) to capture multi-scale temporal dependencies. The bottleneck uses two DPR blocks. The decoder is the mirror version of the encoder, ultimately generating the real and imaginary components of a complex ratio mask (CRM) [20]. This mask is applied to the reference noisy signal through element-wise multiplication to produce the final enhanced spectrum $\hat{S} \in \mathbb{C}^{F \times T}$.

3.4. Loss Fuction

The proposed model is trained using a combination of a spectral magnitude loss, a scale-invariant signal-to-distortion ratio (SI-SDR) [21] loss and a cross-entropy loss of labels. The overall loss function is formulated as:

$$\mathcal{L} = \mathcal{L}_{\text{Mag}} + \lambda \mathcal{L}_{\text{SI-SDR}} + \mathcal{L}_{\text{CE}}, \quad (5)$$

where λ is a weighting factor. The spectral magnitude loss, similar to [22], enforces consistency in the TF domain and is defined as:

$$\mathcal{L}_{\text{Mag}} = \frac{\| |\text{STFT}(\hat{s}_c)| - |\text{STFT}(s_c)| \|_1}{\| |\text{STFT}(s_c)| \|_1}. \quad (6)$$

To improve the perceptual quality of the extracted speech, we incorporate the SI-SDR loss:

$$\mathcal{L}_{\text{SI-SDR}} = - \sum_{c=1}^C 10 \log_{10} \left(\frac{\|s_c\|_2^2}{\|\hat{s}_c - \alpha_c s_c\|_2^2} \right), \quad (7)$$

where α_c is the optimal scaling factor given by

$$\alpha_c = \frac{s_c^T \hat{s}_c}{s_c^T s_c}. \quad (8)$$

$\mathcal{L}_{\text{CE}}$ is the cross-entropy loss between the true state probabilities $p(c_i(n))$ and the predicted probabilities $p(\hat{c}_i(n))$ obtained from the softmax function for the states, DS, FST, NST and DT, $i \in [0, 1, 2, 3]$ corresponding to states 0, 1, 2, and 3, respectively, n is the frame index.

$$\mathcal{L}_{\text{CE}} = - \sum_n \sum_{i=0}^3 p(c_i(n)) \log(p(\hat{c}_i(n))) \quad (9)$$

4. EXPERIMENTS

4.1. Dataset

For training, we generate the simulated data by taking a similar way as reported in [23]. The far-end and near-end signals are chosen from *train-clean-100* subset from the LibriSpeech corpus [24], while the background noise is taken from the DEMAND dataset [25].

To simulate spatialized speech and noise signals, rooms are generated by randomly sampled width, depth, and height from the prescribed intervals of [5, 12] m, [5, 12] m, and [2, 5] m, respectively. The reverberation time (RT60) of the room is randomly varied within the range [0.3, 1.2] s. A circular 4-microphone array is employed as reference configurations with 10 cm diameters. The RIR is generated using *gpuRIR* [26]. For static scenarios, the near-end speaker is positioned at least 0.3 m from walls, while two loudspeakers are systematically distributed in room corners. For moving scenarios, the near-end speaker's start location and moving direction are randomly sampled in the loop trajectories. The moving speed is sampled in [1, 1.5] m/s. Note that, when moving speed is high, the simulated direct-path target signals may have some clicking noise and other audible artifacts. To mitigate this problem, the speech signals at two adjacent locations are partial-overlapped and applied with a trapezium window for both the reverberant signal and direct-path signals [15].

For data augmentation, we simulate realistic acoustic echo cancellation conditions by modeling both nonlinear distortions and time-varying transmission effects. Following [27], we simulate loudspeaker nonlinearities through hard clipping and distortion applied to far-end signals. To model transmission channel variations, we introduce time-varying

delays by applying temporal shifts randomly sampled from $\mathcal{U}(-20, 0)$ ms to each far-end signal, as proposed in [28], which forces continuous adaptation during processing. Our double-talk simulation produces 10-second mixed audio segments containing echo signals and near-end speech, each with randomized durations between 6-10 seconds.

This dataset encompassed signal-to-noise ratio (SNR) ranging from [-10, 10] dB. We mix the noise and the near-end speech based on SNR. The signal-to-echo ratio (SER), which is evaluated during double-talk scenarios, randomly ranges from [-15, 15] dB. We mix the echo signal and the near-end speech based on SER. The Root-Mean-Square (RMS) level of each mixed signal is randomly sampled from [-20, -15] dB. Meanwhile, the signals' energy of the near-end and the echo are calculated and compared to the threshold, judging whether there is voice at the corresponding branch. Four categories are obtained, i.e., Double Silence(DS), Far-end single talk(FST), Near-end single talk(NST), Double Talk(DT), acting as classification labels. All audio mixtures are 10 second-long, with a sampling rate of 16 kHz. In total, we generate 40 hours of training data for static speaker scenarios and an additional 40 hours for moving speaker scenarios. Scenario distribution follows 8:1:1 ratios for DT, FST, and NST conditions.

The test set was simulated using the same methodology as the training set but with different data sources. Clean speech signals were extracted from the *test-clean* subset of LibriSpeech, while noise samples were selected from the same dataset used for training. To evaluate generalization capabilities, we conducted experiments with distinct array geometries: circular, rectangular, and linear configurations. The array size was systematically varied, with radius randomly sampled between 5 cm and 20 cm.

4.2. Experimental Setup

The STFT configuration employs a 16 ms (256 samples) Hanning window with 8 ms (128 samples) hop length, yielding 129 frequency bins. The model architecture adopts $C = 64$ hidden channels and $L = 8$ separation blocks, following [15]'s configuration ($C' = 64$, $C'' = 8$) with channel 0 as reference. The classifier utilizes $D = 32$ hidden dimension, while the PostNet contains 16 hidden channels, resulting in the total parameter count is 0.9M.

The model employs a two-phase training strategy: (1) The base architecture (without PostNet) trains on 4-channel data for 100 epochs to develop robust, array-agnostic features for acoustic echo cancellation; (2) With the base model frozen, the PostNet fine-tunes for 50 epochs to enhance spectral refinement and mitigate AEC performance degradation caused by array configuration differences, effectively addressing residual artifacts and improving speech quality. Training batch size is 4, using the Adam [29] optimizer with an initial learning rate of 0.001, which decays exponentially by a factor of 0.99 per epoch.

Table 1. The performance of the proposed method compared with baselines.

Test scenarios SER(dB)	DT									NST			FST
	-10			-5			0			-			-
	PESQ	SISNRi	WER(%)	PESQ	SISNRi	WER(%)	PESQ	SISNRi	WER(%)	PESQ	SISNRi	WER(%)	ERLE
mix	1.12	-19.42	96.10	1.12	-12.30	91.12	1.14	-8.71	86.01	1.24	-4.70	27.98	-
NKF+MVDR	1.19	9.69	68.37	1.26	6.39	49.40	1.31	3.31	34.10	1.36	0.13	22.14	8.08
ICRN	1.35	17.14	47.22	1.46	13.77	33.48	1.55	10.88	25.09	1.73	7.87	14.94	44.62
Proposed	1.57	21.36	31.13	1.76	17.28	18.44	1.95	14.62	12.68	2.30	11.92	7.46	50.37
w/o Classifier	1.57	21.23	31.30	1.76	17.13	18.89	1.94	14.46	13.05	2.27	11.75	7.58	45.12
w/o Postnet	1.54	20.12	33.40	1.75	17.01	20.23	1.93	12.25	13.70	2.26	11.59	8.15	49.88

Table 2. The performance of the proposed method in static and moving configurations with varying microphone (Q) and loudspeaker (P) counts.

Test scenarios SER(dB)		DT									NST			FST
		-10			-5			0			-			-
near-end	Q\P	PESQ	SISNRi	WER(%)	PESQ	SISNRi	WER(%)	PESQ	SISNRi	WER(%)	PESQ	SISNRi	WER(%)	ERLE
Static	3\1	1.45	18.46	37.43	1.61	15.39	23.34	1.77	12.90	15.79	2.09	10.59	8.90	47.82
	4\1	1.54	19.40	31.51	1.74	16.45	18.75	1.93	14.03	12.60	2.31	11.90	7.10	49.10
	4\3	1.57	20.75	29.35	1.78	17.60	17.30	1.97	14.87	11.89	2.31	11.91	6.94	50.84
	6\3	1.54	20.81	30.96	1.75	17.65	18.55	1.94	14.87	12.80	2.29	11.89	7.86	51.43
Moving	3\2	1.36	16.28	41.17	1.47	12.67	30.24	1.52	9.98	22.16	1.79	7.65	15.43	43.90
	4\2	1.45	17.53	34.04	1.56	14.22	22.18	1.64	12.07	15.05	1.92	8.13	12.50	45.09
	6\2	1.44	17.68	33.87	1.56	14.34	23.45	1.62	12.27	15.57	1.92	8.04	13.22	43.90

4.3. Evaluation Metrics

For evaluation, we employ scenario-specific metrics: SI-SNR improvement (SISNRi) and wideband PESQ [30] for double-talk (DT) and near-end single-talk (NST) conditions, echo return loss enhancement (ERLE) [31] for far-end single-talk (ST-FE). Additionally, we incorporate word error rate (WER) as a downstream automatic speech recognition (ASR) performance metric.

5. RESULTS AND ANALYSIS

5.1. Comparative Evaluation

In Table 1, we present a comprehensive comparison between our proposed model and two baseline methods. The NKF+MVDR, represents a modular approach that combines a neural Kalman filter (NKF) [32] for per-channel echo cancellation with subsequent MVDR beamforming for spatial processing [2]. ICRN [5], implements an end-to-end complex spectral mapping solution that jointly models temporal and spatial dependencies through in-place convolutional recurrent networks for multichannel AEC. For fair comparison, we have scaled the ICRN model size to match our proposed architecture’s parameter count while maintaining its original structural design.

All compared systems are trained on the 4-mic static dataset and evaluated on randomized 4-mic geometries that differ from the training configurations, ensuring the array geometry remains unknown a priori. The test set features varying SNRs uniformly sampled from 5-10 dB, incorporating challenging acoustic conditions such as reverberation, additive noise, and nonlinear distortions. Under these conditions, while the NKF+MVDR approach exhibits limited robustness and the neural-based ICRN framework serves as a strong baseline, our proposed method consistently achieves superior performance across all three acoustic scenarios. As evidenced in Table 1, our method demonstrates significant advantages in processing unseen spatial configurations, outperforming both NKF+MVDR (by 10.8-11.7 dB SISNRi improvement) and ICRN (by 3.5-4.2 dB SISNRi improvement) in cross-geometry evaluation scenarios.

The ablation study systematically quantifies the contribution of each component. By eliminating the classifier and directly feeding the reference microphone signal to the PostNet, we demonstrate that the classifier module plays a crucial role in echo suppression, with its removal leading to a substantial 5.25 dB decrease in ERLE specifically under far-end single-talk conditions. The PostNet, on the other hand, is shown to be principally responsible for perceptual quality enhancement, as indicated by systematic reductions in PESQ scores when deactivated, while concurrently providing a stable 1.0-

2.2% relative improvement in word error rate.

Most notably, in challenging echo-dominant environments with SER of -10 dB, the complete system achieves optimal performance - improving SISNRi by 18.46 dB while dramatically reducing WER from 96.10% to 31.13% compared to the unprocessed baseline. This substantial 64.97% absolute WER reduction underscores the system’s effectiveness in severe acoustic conditions where both reverberation, additive noise, and nonlinear distortions are included.

5.2. Robustness Analysis

5.2.1. Array Configuration Variability

Table 2 demonstrates our model’s generalization capability across different microphone and loudspeaker configurations, despite being trained on a 4-mic/2-loudspeaker setup. When tested with $P = \{1, 3\}$ loudspeakers while maintaining 4 microphones, the model preserves strong acoustic echo cancellation and noise suppression performance, showing remarkable robustness to varying interference conditions. Similarly, adaptation to $Q = \{3, 6\}$ microphone configurations reveals the model’s ability to effectively utilize additional spatial information, with only marginal performance degradation observed in the 6-mic case compared to the original 4-mic setup. The results demonstrate that the model maintains robust generalization capability across different array geometries.

5.2.2. Dynamic Speaker Conditions

Following the same training procedure, we additionally train the model on the moving sound source dataset. Table 2 further evaluates the model’s performance under dynamic conditions with speakers moving at 1-1.5 m/s. Despite the challenging acoustic environment characterized by time-varying spatial relationships, the model demonstrates robust generalization across all tested array configurations ($Q = \{3, 4, 6\}$ mics with $P = 2$ loudspeakers). The results show confirming effective adaptation to both stationary and moving conditions.

5.3. Effectiveness of State-Detection Learning

Figure 2(a) presents a representative example comparing the noisy mixture, far-end signal, target speech, and enhanced output. Figure 2(b) shows the model’s frame-level state predictions alongside the ground truth annotations. The near-perfect alignment between predicted states and reference labels demonstrates the model’s ability to precisely characterize acoustic conditions. This accurate state detection enables optimal processing mode selection for simultaneous AEC and speech enhancement.

The quantitative evaluation on 1,000 test samples demonstrates robust state detection performance, as evidenced by the confusion matrix in Figure 2(c). The system achieves 99% accuracy for critical double-talk (DT) detection-the most com-

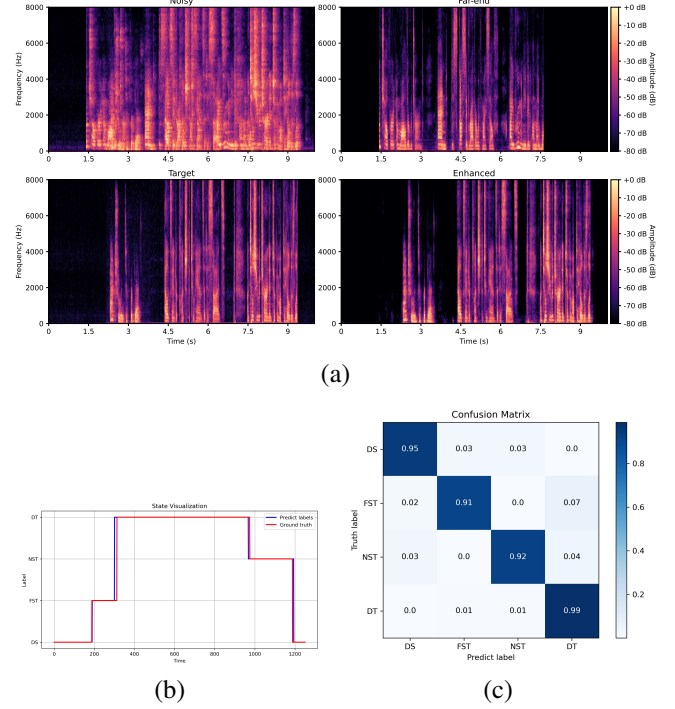


Fig. 2. Experimental results: (a) Spectrum visualizations of noisy input, far-end reference, target speech, and enhanced output signal; (b) Frame-level state transition visualization: ground truth (red) vs predicted (blue); (c) Confusion matrix of state classification (DS: Double Silence, FST: Far-end single talk, NST: Near-end single talk, DT: Double Talk).

putationally demanding state requiring precise echo cancellation, while maintaining high accuracy for double silence (DS, 95%) and near-end/far-end single talk scenarios (NST/FST, 91-92%), the model exhibits a conservative bias toward DT classification.

6. CONCLUSIONS

This work presents a geometry-agnostic neural network for multichannel acoustic echo cancellation, featuring a Dynamic Spatial Encoder for adaptive array processing, joint powerful separation modules for simultaneous echo and noise suppression, and a postnet enhanced by an auxiliary state classifier. Experimental results demonstrate robust performance across diverse microphone configurations, with significant improvements in both speech quality and ASR accuracy, particularly in challenging scenarios involving nonlinear distortions and moving sound sources. The proposed solution provides an effective approach for real-world deployment where array geometries and acoustic conditions vary unpredictably, establishing a new benchmark for flexible AEC systems.

7. REFERENCES

- [1] Gerald Enzner, Herbert Buchner, Alexis Favrot, and Fabian Kuech, “Acoustic echo control,” in *Academic press library in signal processing*, vol. 4, pp. 807–877. Elsevier, 2014.
- [2] Hao Zhang and DeLiang Wang, “A deep learning approach to multi-channel and multi-microphone acoustic echo cancellation,” in *Interspeech*, 2021, pp. 1139–1143.
- [3] Hao Zhang and DeLiang Wang, “Neural cascade architecture for multi-channel acoustic echo suppression,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 2326–2336, 2022.
- [4] Premanand Nayak, Kamini Sabu, and M Shaik, “Multi-mic echo cancellation coalesced with beamforming for real world adverse acoustic conditions,” in *Proc. Interspeech 2024*, 2024, pp. 147–151.
- [5] Chenggang Zhang, Jinjiang Liu, Hao Li, and Xueliang Zhang, “Neural multi-channel and multi-microphone acoustic echo cancellation,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 2181–2192, 2023.
- [6] Yuval Konforti, Israel Cohen, and Baruch Berdugo, “Multichannel acoustic echo cancellation with beamforming in dynamic environments,” *IEEE Open Journal of Signal Processing*, vol. 4, pp. 479–488, 2023.
- [7] Wolfgang Herbordt, Walter Kellermann, and Satoshi Nakamura, “Joint optimization of acoustic echo cancellation and adaptive beamforming,” *Topics in acoustic echo and noise control*, pp. 19–50, 2006.
- [8] Klaus Reindl, Yuanhang Zheng, Anthony Lombard, Andreas Schwarz, and Walter Kellermann, “An acoustic front-end for interactive tv incorporating multichannel acoustic echo cancellation and blind signal extraction,” in *2010 Conference Record of the Forty Fourth Asilomar Conference on Signals, Systems and Computers*. IEEE, 2010, pp. 1716–1720.
- [9] W Herbordt, Satoshi Nakamura, and Walter Kellermann, “Joint optimization of lcmv beamforming and acoustic echo cancellation for automatic speech recognition,” in *Proceedings.(ICASSP’05). IEEE International Conference on Acoustics, Speech, and Signal Processing, 2005*. IEEE, 2005, vol. 3, pp. iii–77.
- [10] Takuya Yoshioka, Xiaofei Wang, Dongmei Wang, Min Tang, Zirun Zhu, Zhuo Chen, and Naoyuki Kanda, “Vararray: Array-geometry-agnostic continuous speech separation,” in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 6027–6031.
- [11] Siyuan Zhang and Xiaofei Li, “Microphone array generalization for multichannel narrowband deep speech enhancement,” in *Proc. Interspeech 2021*, 2021, pp. 666–670.
- [12] Dongheon Lee and Jung-Woo Choi, “Deftan-aa: Array geometry agnostic multichannel speech enhancement,” *Proc. INTERSPEECH, Kos, Greece*, pp. 3360–3364, 2024.
- [13] Zhongweiyang Xu, Xulin Fan, Zhong-Qiu Wang, Xilin Jiang, and Romit Roy Choudhury, “Unsupervised blind speech separation with a diffusion prior,” *arXiv preprint arXiv:2505.05657*, 2025.
- [14] Tianqin Zheng, Jilu Jin, Hanchen Pei, Gongping Huang, Jingdong Chen, and Jacob Benesty, “Spatial-filter-bank-based neural method for multichannel speech enhancement,” *arXiv preprint arXiv:2504.01392*, 2025.
- [15] Changsheng Quan and Xiaofei Li, “Spatialnet: Extensively learning spatial information for multichannel joint speech separation, denoising and dereverberation,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 1310–1323, 2024.
- [16] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton, “Layer normalization,” *ArXiv e-prints*, pp. arXiv–1607, 2016.
- [17] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun, “Delving deep into rectifiers: Surpassing human-level performance on imagenet classification,” in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 1026–1034.
- [18] Haoyin Yan, Jie Zhang, Cunhang Fan, Yeping Zhou, and Peiqi Liu, “Lisennet: Lightweight sub-band and dual-path modeling for real-time speech enhancement,” in *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025, pp. 1–5.
- [19] Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville, “Film: Visual reasoning with a general conditioning layer,” in *Proceedings of the AAAI conference on artificial intelligence*, 2018, vol. 32.
- [20] Donald S Williamson, Yuxuan Wang, and DeLiang Wang, “Complex ratio masking for monaural speech separation,” *IEEE/ACM transactions on audio, speech, and language processing*, vol. 24, no. 3, pp. 483–492, 2015.

- [21] Jonathan Le Roux, Scott Wisdom, Hakan Erdogan, and John R. Hershey, "Sdr – half-baked or well done?," in *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2019, pp. 626–630.
- [22] Zhong-Qiu Wang, Samuele Cornell, Shukjae Choi, Younglo Lee, Byeong-Yeol Kim, and Shinji Watanabe, "Tf-gridnet: Integrating full-and sub-band modeling for speech separation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2023.
- [23] Lu Ma, Song Yang, Yaguang Gong, Xintian Wang, and Zhongqin Wu, "Echofilter: End-to-end neural network for acoustic echo cancellation," *arXiv preprint arXiv:2105.14666*, 2021.
- [24] Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur, "Librispeech: an asr corpus based on public domain audio books," in *2015 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2015, pp. 5206–5210.
- [25] Joachim Thiemann, Nobutaka Ito, and Emmanuel Vincent, "The diverse environments multi-channel acoustic noise database (demand): A database of multichannel environmental noise recordings," in *Proceedings of Meetings on Acoustics*. AIP Publishing, 2013, vol. 19.
- [26] David Diaz-Guerra, Antonio Miguel, and Jose R Beltran, "gpurir: A python library for room impulse response simulation with gpu acceleration," *Multimedia Tools and Applications*, vol. 80, no. 4, pp. 5653–5671, 2021.
- [27] Jung-Hee Kim and Joon-Hyuk Chang, "Attention wave-u-net for acoustic echo cancellation.," in *Interspeech*, 2020, pp. 3969–3973.
- [28] Lukas Pfeifenberger, Matthias Zoehrer, and Franz Pernkopf, "Acoustic echo cancellation with cross-domain learning.," in *Interspeech*, 2021, vol. 8, p. 2021.
- [29] Diederik Kingma and Jimmy Ba, "Adam: A method for stochastic optimization," *Computer Science*, 2014.
- [30] ITUT Rec, "P. 862.2: Wideband extension to recommendation p. 862 for the assessment of wideband telephone networks and speech codecs," *Int. Telecommu. Uni.*, 2005.
- [31] Sergios Theodoridis and Rama Chellappa, *Academic press library in signal processing: Image, video processing and analysis, hardware, audio, acoustic and speech processing*, Academic Press, 2013.
- [32] Dong Yang, Fei Jiang, Wei Wu, Xuefei Fang, and Muyong Cao, "Low-complexity acoustic echo cancellation with neural kalman filtering," in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.